④ Ransomware :- Encrypts the victims files & demands payment to restore success.

⑤ Spyware :- Secretly monitors, browsing activity, credentials & personal info.

27-8-26

→ Case study - 1

* Bright Mart is a medium sized online shopping company. Customers use it's website & mobile app to purchase.

* The company stores

1. Customer names
2. Email addresses
3. Phone no.'s
4. Delivery addresses
5. Passwords
6. Order information
7. Payment related information

- Bright Mart has around 100 employees.
- The company recently hired several new employees. To make things easier, the IT department gave employees to different internal systems.

* Company has the following security weakness:

- Some employees use simple passwords such as BrightMart123.
- Many employees use the same password for multiple systems.
- Software updates are sometimes delayed.
- The company doesn't regularly employee cybersecurity training.
- Theire is no multi-factor authentication for some important accounts.
- Employees can access sensitive info that they don't necessarily need for their jokes.
- Backups are performed only once a month.

* <u>The Incident</u> :-

- One Monday morning, an employee named Rahul receives an email.

<u>Subject</u> :- Urgent - Your account will be suspended.

- The email appears to come from the company's IT department
- It says:

"your company account will be suspended today. Click the below link & verify your password immediately."

- Rahul clicks the link.
- The page looks almost identical to the companies login pages.
- He enters his username & password
- Rahul doesn't realize that the website is controlled by an attacker.
- Now attacker has Rahul's credentials.

* Stage-1 → Unauthorised Access

- Attacker uses Rahul's credentials to access Bright Mart's internal system.
- Because Rahul uses the same password for another system, the attacker is able to access that system too.
- Attacker discovers Rahul has access to customer info.
- Attacker downloads large amount of customer data.

Q1. What cyberscurity problem has occured?

* Stage-2 → Malware Attack

- The attacker sends another email to Rahul containing a malicious attachment.
- Rahul opens the attachment
- Malware is installed on his computer.
- Malware begins communicating with the attacker's system.
- Attacker now begins/attempts to move from Rahul's computer to other systems with Bright Mart's network.

Qd. Which vulnerability allowed the attacker to move further into the organisation?

* Stage-3 → Ransomeware

- Two days later, employees arrive at work & discover that they can't access important files.
- A message appears on their computers:

"Your files have been encrypted. Pay the ransome to recover them."

- The company's customer database & several important business files are encrypted.
- Unfortunately, the company's most recent backup is almost a month old.
- Bright Mart's website stops working
- Customers can't place orders.
- Customer support team can't access customer records

* Stage-4 → Business Impact
  - Bright Mart now faces several problems.
  - Financial Impact
    - The company loses sales because its website is unavailable.
  - Data Impact
    - Customer information may have been stolen.
  - Operational Impact
    - Employees can't access important systems
  - Reputation Impact
    - Customers lose confidence in the company.
  - Legal / Compliance Impact
    - Company may face regulatory or legal consequences depending on what information was exposed & which laws apply.

A1. 1. Phishing
2. Credential theft
3. Unauthorized access
4. Data breach.

A2. Vulnerabilities :-

1. Weak passwords - Reuse of same pass multiple times (diff systems)
2. Excessive user privileges
3. Lack of MFA (Multi-Factor Authentication)
4. Poor network security
5. Lack of employee awareness
6. Outdated software.
7. Delayed backups

Q3) What threats are present in this case?

A) 1. Phishing
2. Malware
3. Ransomware
4. Credential theft
5. Unauthorised access
6. Data theft.

Q4) Was confidential info exposed?

A) Yes, customer info was accessed/downloaded by the attacker

Q5) Was info changed or damaged?

A) Potentially yes, the attacker could modify or encrypt files.

Q6) Who is responsible for cybersecurity?

A) Not only IT department. Every employee who uses the organisation systems has a role in cybersecurity.

Q7) What was the biggest vulnerability?

A) Lack of awareness was the biggest vulnerability because if Rahul knew that if was phishing email, he wouldn't had clicked on the link attached to mail

Q8) Were systems available to legitimate users?

A) The website & internal systems become unavailable because of ransomware.

- Confidentiality - Affected
- Integrity - Affected
- Availability - Affected

Rahul
31/8/26